

Dynamic Linking

Wednesday, October 26, 2022 6:21 PM

- We don't want to make a static library this time, this is because, if we **change our code, the client needs to recompile the code with the new library.** (Note that when we link the static library, it will copy the content in to the executable. So the client needs to recompile in order to get the changes. All the previous commands need to be executed again!)
- Let's create a *.dll.

```
cl /LD /FeMyPrintDynamicLibrary.dll MyPrint.cpp
```

```
10/26/2022 06:25 PM <DIR> .
10/26/2022 06:25 PM <DIR> ..
10/26/2022 06:22 PM      124 MyPrint.cpp
10/26/2022 06:25 PM    77,622 MyPrint.obj
10/26/2022 06:25 PM   174,080 MyPrintDynamicLibrary.dll
10/26/2022 06:25 PM     736 MyPrintDynamicLibrary.exp
10/26/2022 06:25 PM    1,972 MyPrintDynamicLibrary.lib
10/26/2022 05:59 PM      54 Source.cpp
10/26/2022 06:16 PM   178,176 Source.exe
10/26/2022 06:16 PM     707 Source.exp
10/26/2022 06:16 PM    1,708 Source.lib
10/26/2022 06:16 PM     677 Source.obj
      10 File(s)      435,856 bytes
       2 Dir(s)  778,681,815,040 bytes free
```

- This command will generate "MyPrintDynamicLibrary.dll" and "MyPrintDynamicLibrary.lib" files.
- When we compare the static lib and the lib generated above, the size of static lib will be more since the all code resided inside the static lib and this static lib again gets copied into the executable when we do static linking.
- In case of dynamic linking, the code is sitting inside the *.dll. The "MyPrintDynamicLibrary.lib" is kind of intermediate file which will be used during the compilation and linking process. But the address will be resolved dynamically.

```
cl Source.cpp /link MyPrintDynamicLibrary.lib
```

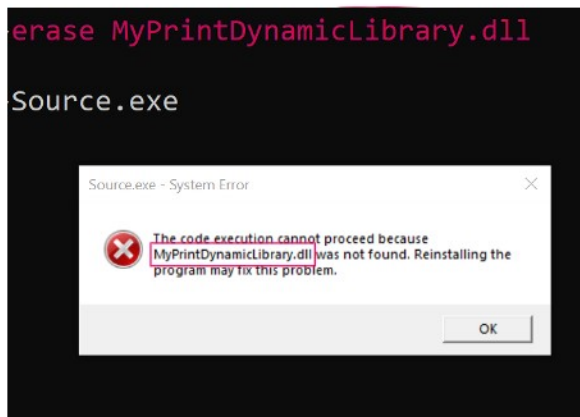
```

10/26/2022 06:25 PM <DIR> .
10/26/2022 06:25 PM <DIR> ..
10/26/2022 06:28 PM      114 MyPrint.cpp
10/26/2022 06:28 PM    77,612 MyPrint.obj
10/26/2022 06:28 PM   174,080 MyPrintDynamicLibrary.dll → Size is more
10/26/2022 06:28 PM     736 MyPrintDynamicLibrary.exp
10/26/2022 06:25 PM   1,972 MyPrintDynamicLibrary.lib → Size is less
10/26/2022 05:59 PM     54 Source.cpp
10/26/2022 06:29 PM   76,288 Source.exe
10/26/2022 06:16 PM     707 Source.exp
10/26/2022 06:16 PM    1,708 Source.lib
10/26/2022 06:29 PM     677 Source.obj
          10 File(s)      333,948 bytes
          2 Dir(s)  778,674,049,024 bytes free

C:\Users\Nishith-Jain.MR\source\repos\DynamicLinking>Source.exe
Hello

```

- Now we don't require "MyPrintDynamicLibrary.lib" file.
- This is because, the code is dynamically linked using *.dll.
- Now if we erase the "MyPrintDynamicLibrary.dll" and run the "Source.exe", then we get a run-time error telling "MyPrintDynamicLibrary.dll" is missing on computer.



- Now we can change the source code of dll (MyPrint.cpp) and this doesn't require the Source.cpp to recompile to get the new changes as it is linked dynamically!
- Let's change the `Hello()` method in the "MyPrint.cpp".

```

#include <iostream>
using namespace std;

__declspec(dllexport) void Hello()
{
    cout << "Hello My World!" << endl;
}

```

- Now we have to just recompile the dll!

```

cl /LD /FeMyPrintDynamicLibrary.dll MyPrint.cpp

```

- Now just run the executable Source.exe

```
10/26/2022 06:36 PM <DIR> .
10/26/2022 06:36 PM <DIR> ..
10/26/2022 06:56 PM      124 MyPrint.cpp
10/26/2022 06:57 PM    77,622 MyPrint.obj
10/26/2022 06:57 PM    174,080 MyPrintDynamicLibrary.dll
10/26/2022 06:57 PM      736 MyPrintDynamicLibrary.exp
10/26/2022 06:25 PM     1,972 MyPrintDynamicLibrary.lib
10/26/2022 05:59 PM       54 Source.cpp
10/26/2022 06:37 PM    76,288 Source.exe
10/26/2022 06:16 PM       707 Source.exp
10/26/2022 06:16 PM     1,708 Source.lib
10/26/2022 06:37 PM       677 Source.obj
        10 File(s)      333,968 bytes
         2 Dir(s)  778,664,017,920 bytes free
```

```
C:\Users\Nishith-Jain.MR\source\repos\DynamicLinking>Source.exe
Hello My World!
```

Observe the
time stamp.